5-2c Purchases Transactions

There are two systems for accounting for merchandise transactions: perpetual and periodic. In a **perpetual inventory system** (The inventory system in which each purchase and sale of merchandise is recorded in the inventory account and related subsidiary ledger; therefore, the inventory records are updated continuously.), each purchase and sale of merchandise is recorded in the inventory account and related subsidiary ledger. In this way, the amount of merchandise available for sale and the amount sold are continuously (perpetually) updated in the inventory records. In a **periodic inventory system** (An inventory system in which the inventory records are updated only after a physical count has been taken at periodic intervals, usually at the end of an accounting period.), the inventory does not show the amount of merchandise available for sale and the amount sold. Instead, a listing of inventory on hand, called a **physical inventory** (A detailed listing of merchandise on hand.), is prepared at the end of the accounting period. This physical inventory is used to determine the cost of inventory on hand at the end of the period and the cost of goods sold during the period.

Most retail companies use computerized perpetual inventory systems. Such systems use bar codes or radio frequency identification codes embedded in a product. An optical scanner or radio frequency identification device is then used to read the product codes and track inventory on hand and sold.

Because computerized perpetual inventory systems are widely used, this chapter illustrates merchandise transactions using a perpetual inventory system. The periodic system is described and illustrated in Appendix 2 at the end of this chapter.

Link to Dollar Tree

Dollar Tree uses point-of-sale computerized software to plan purchases and track inventory. This system automatically reorders key items based on sales and inventory levels.

Under the perpetual inventory system, cash purchases of merchandise are recorded as follows:

To illustrate the impact of each journal entry on the accounting equation and the financial statement elements, T account postings are shown after each journal entry. The T account postings for the preceding journal entry are as follows:

Purchases of inventory on account are recorded as follows:

Jan. 4	Inventory	9,250	
	Accounts Payable—Thomas Corporation		9,250
	Purchased inventory on account.		

Assets	=	Liabilities	+	Stockholders' Equity
Inventory		Accounts Payable		
9,250 |		| 9,250		

The terms of purchases on account are normally indicated on the **invoice** (The bill that the seller sends to the buyer for sales made or services rendered on account.) or bill that the seller sends the buyer. An example of an invoice sent to **NetSolutions** by Alpha Technologies is shown in Exhibit 3.

Exhibit 3

Invoice

Alpha Technologies
1000 Matrix Blvd.
San Jose, CA 95116-1000

Invoice
106-8

Made in U.S.A.

SOLD TO	CUSTOMER ORDER NO.	ORDER DATE	
NetSolutions 5101 Washington Ave. Cincinnati, OH 45227-5101	412	Jan. 3, 20Y8	

DATE SHIPPED	HOW SHIPPED AND ROUTE	TERMS	INVOICE DATE
Jan. 5, 20Y8	US Express Trucking Co.	2/10, n/30	Jan. 5, 20Y8
FROM	**F.O.B.**		
San Jose	Cincinnati		

QUANTITY	DESCRIPTION	UNIT PRICE	AMOUNT
20	HC9 Printer/Fax/Copiers	150.00	3,000.00

The terms for when payments for merchandise are to be made are called the **credit terms** (Terms for payment on account by the buyer to the seller.) . If payment is required on delivery, the terms are cash or net cash. Otherwise, the buyer is allowed an amount of time, known as the **credit period** (The amount of time the buyer is allowed in which to pay the seller.) , in which to pay. The credit period usually begins with the date of the sale as shown on the invoice.

Business Insight

Apple's Credit Terms

Working capital efficiency is influenced by the relationship between the suppliers' and customers' credit terms. If the suppliers' credit terms are longer than the customers' credit terms, then the company is able to use suppliers to finance the operating cycle. For example, **Apple (AAPL)** is able to collect on sales within an average of approximately 30 days. However, Apple uses an average of approximately 100 days to pay its suppliers. Thus, Apple collects faster than it pays, allowing Apple to use the suppliers' money as an interest-free loan for the 70-day (100 days – 30 days) difference.

Source: *Apple*®.

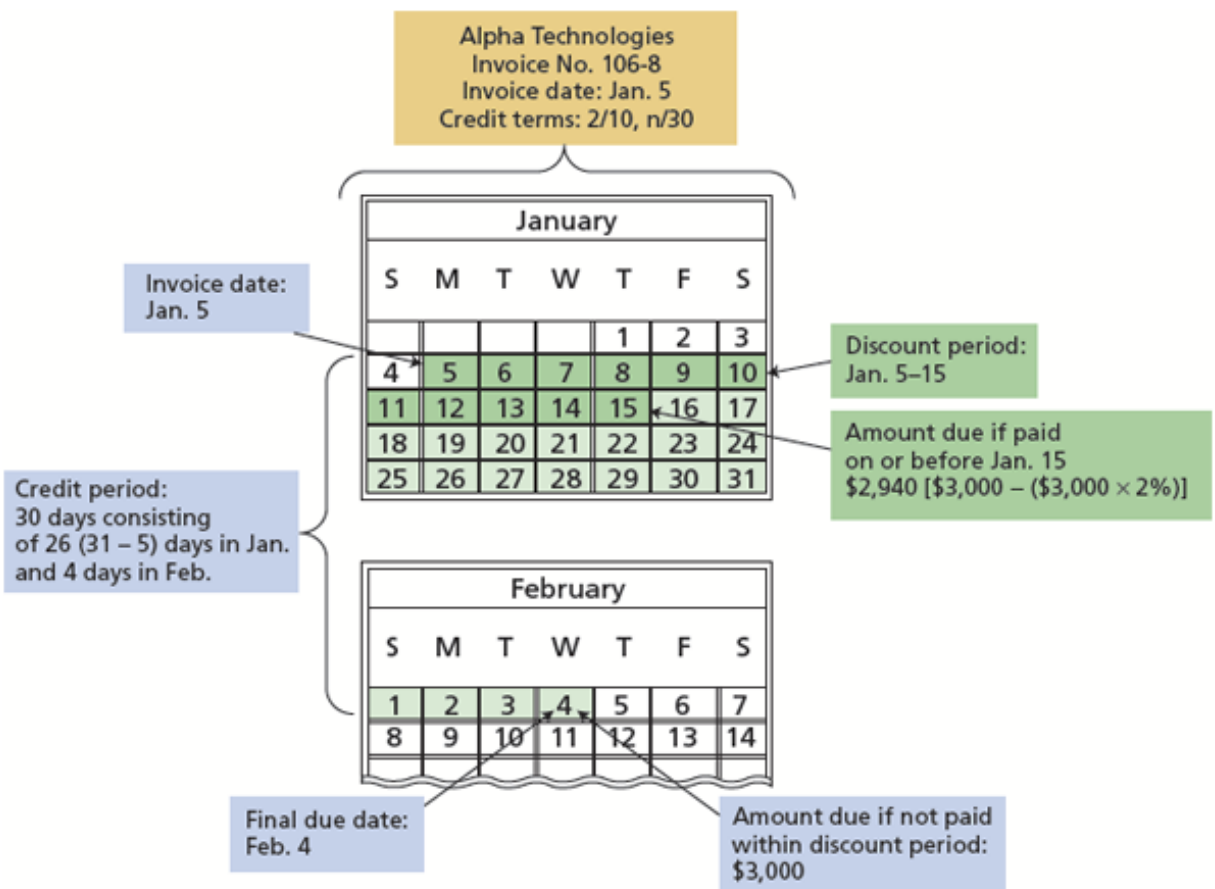
Payment may be due within a stated number of days after the invoice date, such as 15 days or 30 days. In these cases, the terms are net 15 days or net 30 days and are normally expressed as n/15 or n/30. If payment is due by the end of the month in which the sale was made, the terms are written as *n/eom*.

Purchases Discounts

B2B (business-to-business) purchases may include a discount by the supplier (vendor) to encourage the buyer to pay before the end of the credit period. For example, a supplier may offer a 2% discount if the buyer pays within 10 days of the invoice date. If the buyer does not take the discount, the total invoice amount is due within 30 days. These terms are expressed as 2/10, n/30 and are read as "2% discount if paid within 10 days, net amount due within 30 days." The credit terms of 2/10, n/30 are summarized in Exhibit 4, using the invoice in Exhibit 3.

Exhibit 4

Credit Terms



Discounts taken by the buyer for early payment of an invoice are called **purchases discounts** (Discounts taken by the buyer for early payment of an invoice.) .

Purchases discounts taken by a buyer reduce the cost of the merchandise purchased. Even if the buyer has to borrow to pay within a discount period, it is normally to the buyer's advantage to do so. For this reason, accounting systems are normally designed so that all available discounts are taken.

To illustrate, the invoice shown in Exhibit 3 is used. The last day of the discount period is January 15 (invoice date of January 5 plus 10 days). Assume that in order to pay the invoice on January 15, **NetSolutions** borrows $2,940, which is $3,000 less the discount of $60 ($3,000 × 2%). If an annual interest rate of 6% and a 360-day year is also assumed, the interest on the loan of $2,940 for the remaining 20 days of the credit period is $9.80 ($2,940 × 6% × 20 ÷ 360).

The net savings to NetSolutions of taking the discount is $50.20, computed as follows:

Discount of 2% on $3,000	$60.00
Interest for 20 days at a rate of 6% on $2,940	(9.80)
Savings from taking the discount	$50.20

The savings can also be seen by comparing the interest rate on the money *saved* by taking the discount and the interest rate on the money *borrowed* to take the discount. The interest rate on the money saved in the prior example is estimated by converting 2% for 20 days to a yearly rate, as follows:

$$2\% \times \frac{360\text{ days}}{20\text{ days}} = 2\% \times 18 = 36\%$$

If NetSolutions borrows the money to take the discount, it pays interest of 6%. If NetSolutions does not take the discount, it *pays* an estimated interest rate of 36% for using the $3,000 for an additional 20 days.

Under the perpetual inventory system, the buyer initially debits Inventory for the amount of the invoice. When paying the invoice within the discount period, the buyer credits Inventory for the amount of the discount. In this way, Inventory shows the *net* cost to the buyer.

To illustrate, NetSolutions would record the Alpha Technologies invoice and its payment at the end of the discount period as follows:

Jan.	5	Inventory	3,000	
		Accounts Payable—Alpha Technologies		3,000

Assets	=	Liabilities	+	Stockholders' Equity
Inventory: 3,000 (debit)		Accounts Payable: 3,000 (credit)		

Jan.	15	Accounts Payable—Alpha Technologies	3,000	
		Cash		2,940
		Inventory		60

Assets	=	Liabilities	+	Stockholders' Equity
Cash: 2,940 (credit); Inventory: 60 (credit)		Accounts Payable: 3,000 (debit)		

Assume that NetSolutions does not take the discount, but instead pays the invoice on February 4. In this case, NetSolutions would record the payment as follows:

Feb.	4	Accounts Payable—Alpha Technologies	3,000	
		Cash		3,000

Assets	=	Liabilities	+	Stockholders' Equity
Cash		Accounts Payable		
(credit) 3,000		(debit) 3,000		

Purchases Returns and Allowances

A buyer may request an allowance for merchandise that is returned (purchases return) or a price allowance (purchases allowance) for damaged or defective merchandise. From a buyer's perspective, such returns and allowances are called
purchases returns and allowances (From the buyer's perspective, returned merchandise or an adjustment for defective merchandise.)
. In both cases, the buyer normally sends the seller a debit memorandum to notify the seller of reasons for the return (purchase return) or to request a price reduction (purchase allowance).

A
debit memorandum (A form used by a buyer to inform the seller of the amount the buyer proposes to debit to the account payable due the seller.)
, often called a
debit memo (A form used by a buyer to inform the seller of the amount the buyer proposes to debit to the account payable due the seller.)
, is shown in Exhibit 5. A debit memo informs the seller of the amount the buyer proposes to *debit* to the account payable due the seller. It also states the reasons for the return or the request for the price allowance.

Exhibit 5

Debit Memo

NetSolutions
5101 Washington Ave.
Cincinnati, OH 45227-5101

No. 18

DEBIT MEMO

TO
Maxim Systems
7519 East Wilson Ave.
Seattle, WA 98101-7519

DATE
March 7, 20Y8

WE DEBITED YOUR ACCOUNT AS FOLLOWS

10	Server Network Cards, your invoice No. 7291, are being returned via parcel post. Our order specified No. 825X.	@90.00	900.00

The buyer may use the debit memo as the basis for recording the return or allowance or wait for approval from the seller (creditor). In either case, the buyer debits Accounts Payable and credits Inventory.

To illustrate, NetSolutions records the return of the inventory indicated in the debit memo in

Exhibit 5 as follows:

Mar. 7	Accounts Payable—Maxim Systems	900	
	Inventory		900
	Debit Memo No. 18.		

Assets	=	Liabilities	+	Stockholders' Equity
Inventory		Accounts Payable		
Cr. 900		Dr. 900		

Before paying an invoice, a buyer may return inventory or be granted a price allowance for an invoice with a purchase discount. To illustrate, assume the following data concerning a purchase of inventory by NetSolutions on May 2:

May 2. Purchased $5,000 of inventory on account from Delta Data Link, terms 2/10, n/30.

4. Returned inventory with an invoice amount of $1,000 that was purchased on May 2.

12. Paid for the purchase of May 2 less the return and discount.

NetSolutions would record these transactions as follows:

May 2	Inventory	5,000	
	Accounts Payable—Delta Data Link		5,000
	Purchased merchandise.		

Assets	=	Liabilities	+	Stockholders' Equity
Inventory		Accounts Payable		
Dr. 5,000		Cr. 5,000		

May 4	Accounts Payable—Delta Data Link	1,000	
	Inventory		1,000
	Returned portion of merchandise purchased.		

Assets	=	Liabilities	+	Stockholders' Equity
Inventory		Accounts Payable		
Cr. 1,000		Dr. 1,000		

May 12	Accounts Payable—Delta Data Link	4,000	
	Inventory [($5,000 – $1,000) × 2%]		80
	Cash ($4,000 – $80)		3,920
	Paid invoice within discount period.		

Assets		=	Liabilities	+	Stockholders' Equity
Cash	Inventory		Accounts Payable		
Cr. 3,920	Cr. 80		Dr. 4,000		

Check up Corner 5-1

Purchases Transactions

On August 10, Knopfler Company purchased inventory on account from Fray Co. for $11,500, terms 2/10, n/30. Knopfler returned inventory with an invoice amount of $2,500 on August 14 and received full credit. Journalize Knopfler's entries for (a) the purchase, (b) the return, and (c) the payment of the invoice on August 18.

Pathways Challenge

This Is Accounting!

Economic Activity

To make sure that they have an adequate supply of inventory, many companies enter into contracts to purchase merchandise and other inventory at a future date. For example, in recent financial statements, **Starbucks Corporation (SBUX)** reported that it had entered into $854 million of contracts (commitments) to purchase coffee in the future at a fixed price. It had another $203 million of similar contracts (commitments) where the price was to be determined at a later date.

Source: Starbucks Corporation, *Form 10-K for a Recent Fiscal Year Ended September 29*.

Critical Thinking/Judgment

How should Starbucks record and/or report these contracts (commitments) to purchase coffee?

If the market price of coffee falls below the agreed-upon fixed price, should Starbucks recognize a loss?

If the market price of coffee rises above the agreed-upon fixed price, should Starbucks recognize a gain?